Pattern Recognition and Machine Learning

Chapter 3: Linear Models for Regression

Reading Notes

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Linear Basis Function Models

Simplest Linear Model:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + \cdots + w_D x_D$$

Linear Model with Basis Functions:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathbf{x})$$

Vector Form (with $\phi_0(\mathbf{x}) = 1$):

$$y(\mathbf{x}, \mathbf{w}) = \sum_{j=0}^{M-1} w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$$

Basis Functions:

- **Polynomial**: $\phi_j(x) = x^j$
- **Gaussian**: $\phi_j(x) = \exp\left(-\frac{(x-\mu_j)^2}{2s^2}\right)$
- **Sigmoidal**: $\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right)$

Maximum Likelihood and Least Squares

Probabilistic Model:

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \beta^{-1})$$

Likelihood Function:

$$p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}(t_n|\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n), \beta^{-1})$$

Log-Likelihood:

$$\ln p(\mathbf{t}|\mathbf{w}, \beta) = \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi) - \beta E_D(\mathbf{w})$$

$$E_D(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n)\right)^2$$

Maximizing the likelihood is equivalent to minimizing the sum-of-squares error.

Maximum Likelihood Solution

$$E_D(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)^2$$

$$\nabla_{\mathbf{w}} E_D(\mathbf{w}) = -\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)\phi(\mathbf{x}_n)$$

$$\text{Set to zero: } 0 = -\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)\phi(\mathbf{x}_n)$$

$$\sum_{n=1}^{N} t_n\phi(\mathbf{x}_n) = \sum_{n=1}^{N}\phi(\mathbf{x}_n)\phi(\mathbf{x}_n)^T\mathbf{w}$$

$$\mathbf{\Phi}^T\mathbf{t} = \mathbf{\Phi}^T\mathbf{\Phi}\mathbf{w}$$

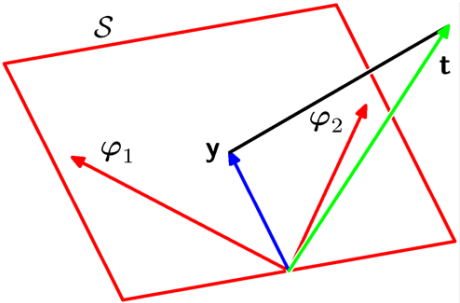
$$\mathbf{w}_{ML} = (\mathbf{\Phi}^T\mathbf{\Phi})^{-1}\mathbf{\Phi}^T\mathbf{t}$$

The design matrix $\mathbf{\Phi}$ is defined as:

$$\mathbf{\Phi} = \begin{bmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_{M-1}(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{bmatrix}$$

- $\phi_0(\mathbf{x}) = 1$ (bias term)
- $\phi_j(\mathbf{x})$: basis functions
- M: number of parameters
- N: number of data points

Geometry of Least Squares



- $\mathbf{t}$: Target vector in N-dimensional space
- φ_j: Basis function vectors
- $\mathbf{y}$: Prediction vector in subspace S

Geometric Meaning of Normal Equations

The condition $\mathbf{\Phi}^T(\mathbf{t} - \mathbf{\Phi w}) = \mathbf{0}$ means the residual vector is orthogonal to all basis functions, ensuring optimal projection onto the model space.

Sequential Learning

Stochastic Gradient Descent:

Parameter Update:

$$\begin{aligned}\mathbf{w}^{(\tau+1)} &= \mathbf{w}^{(\tau)} - \eta \nabla E_n(\mathbf{w}) \\ &= \mathbf{w}^{(\tau)} + \eta(t_n - \mathbf{w}^{(\tau)T}\phi_n)\phi_n\end{aligned}$$

- Also known as **LMS algorithm**
- Processes one data point at a time
- Suitable for large datasets or online learning
- Learning rate η requires careful tuning

Advantage

Avoids storing entire dataset in memory - useful for big data applications

Regularized Least Squares

Regularized Error Function:

$$E(\mathbf{w}) = E_D(\mathbf{w}) + \lambda E_W(\mathbf{w})$$

Weight Decay (L_2 regularization):

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)^2 + \frac{\lambda}{2}\mathbf{w}^T\mathbf{w}$$

Closed-form Solution:

$$\mathbf{w} = (\lambda\mathbf{I} + \mathbf{\Phi}^T\mathbf{\Phi})^{-1}\mathbf{\Phi}^T\mathbf{t}$$

Bayesian Interpretation

Equivalent to MAP estimation with Gaussian prior: $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$
where $\lambda = \alpha/\beta$ relates prior precision α to noise precision β

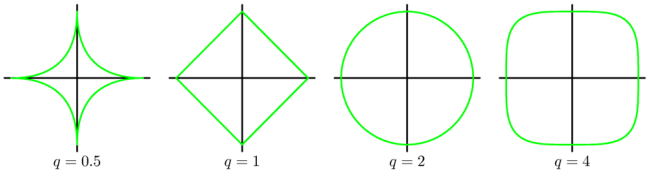
General Regularization

General L_q Regularization:

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{M}|w_j|^q$$

Common Cases:

- $q = 1$: **Lasso** - sparse solutions
- $q = 2$: **Ridge** - shrink coefficients
- $q < 1$: Non-convex regularization



Sparsity Property

Lasso ($q = 1$) tends to produce sparse models with many zero weights

The Bias-Variance Decomposition

Expected Squared Loss Decomposition:

$$\mathbb{E}[(y(\mathbf{x}) - t)^2] = \underbrace{(\mathbb{E}[y(\mathbf{x})] - \mathbb{E}[t|\mathbf{x}])^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(y(\mathbf{x}) - \mathbb{E}[y(\mathbf{x})])^2]}_{\text{Variance}} + \underbrace{\mathbb{E}[(t - \mathbb{E}[t|\mathbf{x}])^2]}_{\text{Noise}}$$

Intuitive Interpretation:

- **Bias**: Difference between average prediction and true conditional expectation
- **Variance**: Variability of predictions across different datasets
- **Noise**: Irreducible uncertainty in the data

Practical Implications:

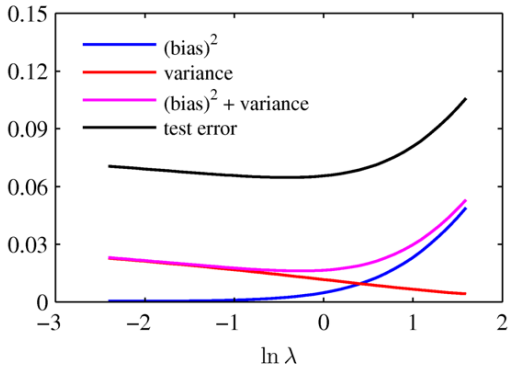
- Simple models: Low variance, High bias
- Complex models: High variance, Low bias
- Goal: Find optimal trade-off

Model Complexity Trade-off

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

Regularization helps control this trade-off by adjusting model complexity

Bias-Variance Trade-off



- **Small λ (Weak regularization)**:
  - Low bias, high variance
  - Overfitting regime
  - Complex model fits noise
- **Large λ (Strong regularization)**:
  - High bias, low variance
  - Underfitting regime
  - Oversimplified model
- **Optimal λ**:
  - Balances bias and variance
  - Minimizes total error

Finding the Sweet Spot

The optimal regularization strength λ^* minimizes the sum of bias^2 and variance

Conjugate Priors: Key Bayesian Concept

Definition: Prior is conjugate to likelihood if posterior belongs to same family.

Bayesian Update:

$$\underbrace{p(\boldsymbol{\theta}|\mathcal{D})}_{\text{Posterior}} \propto \underbrace{p(\mathcal{D}|\boldsymbol{\theta})}_{\text{Likelihood}} \times \underbrace{p(\boldsymbol{\theta})}_{\text{Prior}}$$

Example: Gaussian-Gaussian

- Prior: $\mathcal{N}(\mu_0, \sigma_0^2)$
- Likelihood: $\mathcal{N}(\mu, \sigma^2)$
- Posterior: Gaussian

Advantages:

- Analytical solutions
- Efficient computation
- Interpretable updates

In Bayesian Linear Regression

Gaussian prior + Gaussian likelihood $\rightarrow$ Gaussian posterior

Enables exact closed-form solutions for posterior and predictive distributions

Bayesian Linear Regression

Prior Distribution:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0)$$

Posterior Distribution:

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_N, \mathbf{S}_N)$$

$$\begin{aligned}
\ln p(\mathbf{w}|\mathbf{t}) &= \ln p(\mathbf{t}|\mathbf{w}) + \ln p(\mathbf{w}) + \text{const} \\
&= -\frac{\beta}{2}\sum_{n=1}^{N}(t_n - \mathbf{w}^T\phi_n)^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^T\mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) + \text{const} \\
&= -\frac{1}{2}\mathbf{w}^T(\beta\mathbf{\Phi}^T\mathbf{\Phi} + \mathbf{S}_0^{-1})\mathbf{w} + \mathbf{w}^T(\beta\mathbf{\Phi}^T\mathbf{t} + \mathbf{S}_0^{-1}\mathbf{m}_0) + \text{const}
\end{aligned}$$

$$\ln p(\mathbf{w}|\mathbf{t}) = \ln \mathcal{N}(\mathbf{w}|\mathbf{m}_N, \mathbf{S}_N) = -\frac{1}{2}\mathbf{w}^T\mathbf{S}_N^{-1}\mathbf{w} + \mathbf{w}^T\mathbf{S}_N^{-1}\mathbf{m}_N + \text{const}$$

Posterior Parameters:

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \beta\mathbf{\Phi}^T\mathbf{\Phi}, \quad \mathbf{m}_N = \mathbf{S}_N(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\mathbf{\Phi}^T\mathbf{t})$$

Predictive Distribution

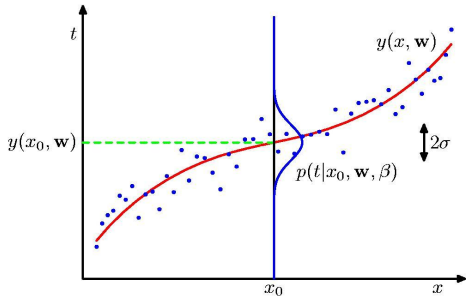
Predictive Distribution:

$$p(t|\mathbf{x}, \mathbf{t}, \alpha, \beta) = \int p(t|\mathbf{x}, \mathbf{w}, \beta) p(\mathbf{w}|\mathbf{t}, \alpha, \beta) d\mathbf{w}$$

Predictive Distribution Form:

$$p(t|\mathbf{x}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(t|\mathbf{m}_N^T \phi(\mathbf{x}), \sigma_N^2(\mathbf{x}))$$

$$\sigma_N^2(\mathbf{x}) = \frac{1}{\beta} + \phi(\mathbf{x})^T \mathbf{S}_N \phi(\mathbf{x})$$

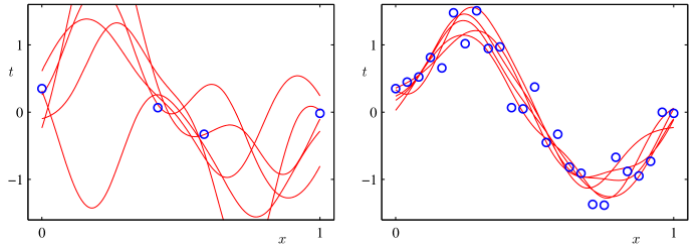


- Blue curve: conditional distribution $p(t|x)$ at a specific input x_0

Practical Uncertainty Quantification

The predictive distribution answers: "Given the data, what is the probability that the target value falls within a certain range at this input location?"

Bayesian Predictive Distribution



Posterior samples of $y(\mathbf{x}; \mathbf{w})$: $N = 4$ (left) and $N = 25$ (right).

- **Sample functions**: one curve per $\mathbf{w}$ sampled from the posterior.
- Their spread shows the covariance between the predictions at different $\mathbf{x}$.
- More data concentrates the curves: the posterior over $\mathbf{w}$ tightens.

Bayesian regression delivers a distribution over functions, not a single fitted curve.

Bayesian Model Comparison

Model Posterior:

$$p(\mathcal{M}_i|\mathcal{D}) \propto p(\mathcal{D}|\mathcal{M}_i)\, p(\mathcal{M}_i)$$

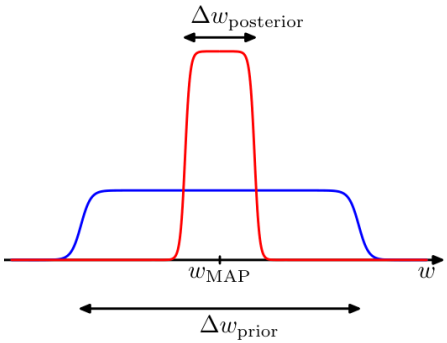
Evidence:

$$p(\mathcal{D}|\mathcal{M}_i) = \int p(\mathcal{D}|\mathbf{w}, \mathcal{M}_i)\, p(\mathbf{w}|\mathcal{M}_i)\, \mathrm{d}\mathbf{w}$$

$$p(\mathcal{D}) \simeq p(\mathcal{D}|\mathbf{w}_{\mathrm{MAP}})\, \frac{\Delta \mathbf{w}_{\mathrm{posterior}}}{\Delta \mathbf{w}_{\mathrm{prior}}}$$

Log Evidence (M parameters):

$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|\mathbf{w}_{\mathrm{MAP}}) + M \ln \left(\frac{\Delta w_{\mathrm{posterior}}}{\Delta w_{\mathrm{prior}}} \right)$$



Evidence trades fit against posterior volume: complexity is penalized without a validation set.

Evidence Approximation: Motivation and Method

The Challenge with Full Bayesian Treatment:

- Ideally, we want to integrate out all parameters:

$$p(t|\mathbf{t}) = \iiint p(t|\mathbf{w},\beta)p(\mathbf{w}|\mathbf{t},\alpha,\beta)p(\alpha,\beta|\mathbf{t})d\mathbf{w}d\alpha d\beta$$

- This triple integral is analytically intractable
- Numerical integration is computationally expensive

Evidence Approximation Solution:

- Approximate $p(\alpha,\beta|\mathbf{t})$ by point estimates $\hat{\alpha},\hat{\beta}$
- Find optimal hyperparameters by maximizing marginal likelihood:

$$\hat{\alpha},\hat{\beta} = \arg\max_{\alpha,\beta} p(\mathbf{t}|\alpha,\beta)$$

- Use these point estimates for prediction:

$$p(t|\mathbf{t}) \approx p(t|\mathbf{t},\hat{\alpha},\hat{\beta}) = \int p(t|\mathbf{w},\hat{\beta})p(\mathbf{w}|\mathbf{t},\hat{\alpha},\hat{\beta})d\mathbf{w}$$

Evidence Maximization: Finding Optimal Hyperparameters

Marginal Likelihood (Evidence):

$$p(\mathbf{t}|\alpha,\beta) = \int p(\mathbf{t}|\mathbf{w},\beta)p(\mathbf{w}|\alpha)d\mathbf{w}$$

Why Maximize Evidence?

The evidence $p(\mathbf{t}|\alpha,\beta)$ measures how likely the data is under the model with given hyperparameters α, β.
Maximizing evidence = Finding hyperparameters that best explain the observed data.

Log Evidence for Linear Regression:

$$\ln p(\mathbf{t}|\alpha,\beta) = \frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - E(\mathbf{m}_N) + \frac{1}{2}\ln|\mathbf{S}_N| - \frac{N}{2}\ln(2\pi)$$

where: $\mathbf{S}_N^{-1} = \alpha\mathbf{I} + \beta\mathbf{\Phi}^T\mathbf{\Phi}$, $\mathbf{m}_N = \beta\mathbf{S}_N\mathbf{\Phi}^T\mathbf{t}$, $E(\mathbf{m}_N) = \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{m}_N\|^2 + \frac{\alpha}{2}\mathbf{m}_N^T\mathbf{m}_N$

Fixed Point Equations and Iterative Solution

Optimization Approach:

- Set derivatives to zero: $\frac{\partial}{\partial \alpha} \ln p(\mathbf{t}|\alpha, \beta) = 0$, $\frac{\partial}{\partial \beta} \ln p(\mathbf{t}|\alpha, \beta) = 0$
- Leads to fixed point equations that must be solved iteratively

Fixed Point Equations:

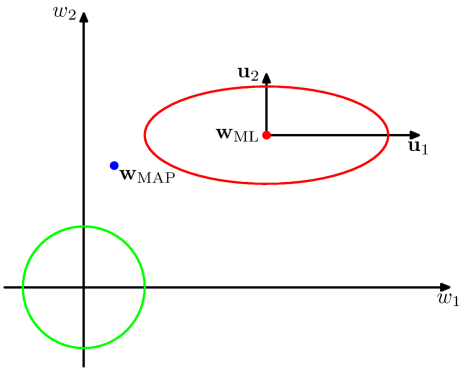
$$\hat{\alpha} = \frac{\gamma}{\mathbf{m}_N^T \mathbf{m}_N}$$

$$\frac{1}{\hat{\beta}} = \frac{1}{N - \gamma} \sum_{n=1}^{N} \left(t_n - \mathbf{m}_N^T \boldsymbol{\phi}(\mathbf{x}_n) \right)^2$$

Practical Significance

This iterative process automatically determines optimal regularization strength, balancing model complexity with data fitting - no manual tuning required!

Effective Parameters and Model Complexity



Effective Parameters:

$$\gamma = \sum_{i=1}^{M} \frac{\lambda_i}{\alpha + \lambda_i}$$

Key Insight:

- λ_i: Data strength along eigenvector i
- α: Prior strength
- $\lambda_i \gg \alpha$: Parameter well-determined by data
- $\lambda_i \ll \alpha$: Parameter dominated by prior

Evidence-Based Complexity Control

The evidence framework automatically balances data fitting against model complexity by comparing eigenvalues λ_i (data evidence) with prior precision α

Limitations of Fixed Basis Functions

Key Limitation:

- **Curse of dimensionality**: M basis functions along each dimension of a D-dimensional input space requires M^D
- **Rigid structure**: fixed before the data are seen, so they cannot adapt to it

Two Properties of Real Data:

- Data lie near a low-dimensional manifold, so localized basis functions need only cover the region holding data
- Targets depend on few directions within that manifold, so the basis functions need only respond to those directions

These limitations motivate the more flexible models of later chapters.

Advanced Approaches in Later Chapters

- Neural networks: adaptive basis functions (Chapter 5)
- Kernel methods: implicit feature spaces (Chapter 6)

Summary

- **Linear Basis Models**: Flexible framework for regression with nonlinear basis functions
- **Maximum Likelihood**: Equivalent to least squares under Gaussian noise assumption
- **Bias-Variance Trade-off**: Fundamental concept in model selection
- **Bayesian Approach**: Natural framework for regularization and uncertainty quantification
- **Evidence Approximation**: Empirical Bayes method for hyperparameter estimation
- **Predictive Distributions**: Provide complete uncertainty information for predictions

Key Contribution

Bayesian linear regression provides a principled probabilistic framework that naturally handles regularization and uncertainty quantification, setting the stage for more advanced methods.